

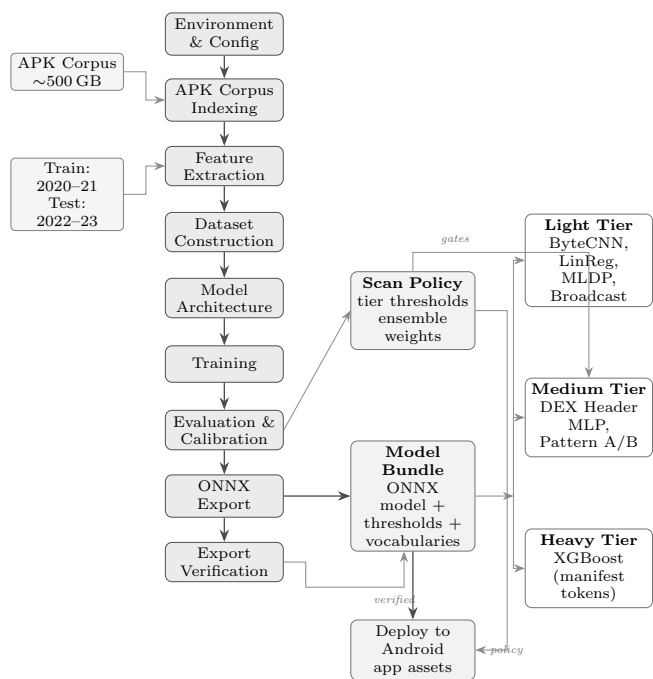


Fig. 1. Offline model-development pipeline. APKs are indexed, features extracted, models trained with temporal splits, and exported as verified ONNX bundles grouped into Light, Medium, and Heavy inference tiers.

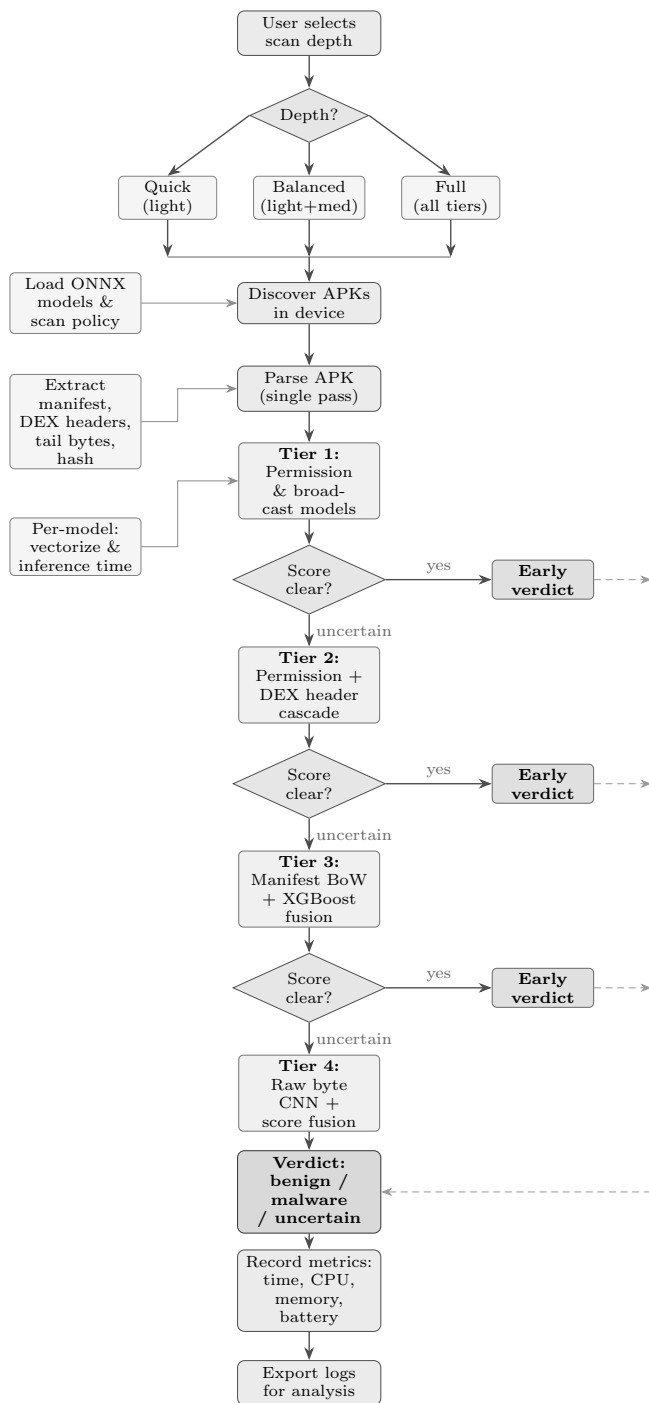


Fig. 2. VigiDroid on-device scanning flow. The app parses each APK once, then runs a tiered cascade of increasingly expensive models—exiting early when a confident score is reached—and logs per-stage resource metrics for thesis evaluation.